

UNIT III - COMPUTER NETWORKS

1. Data-Link Layer (Wireless Networks)

a. Introduction

- The Data-Link Layer is the second layer in the OSI model.
- It is responsible for:
 - **Node-to-node delivery**  (Ek node se dusre tak data bhejna)
 - **Framing**  (Data ko chhote packets me todna)
 - **Error detection and correction** 
 - **MAC (Media Access Control)**

b. IEEE 802.11 (Wi-Fi)

- Standards for **wireless local area networks (WLAN)** 
- Frequency bands: **2.4 GHz & 5 GHz**
- Uses **CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)**  
- Modes: Infrastructure (with access point) & Ad-hoc (device-to-device)
- Security protocols: WEP, WPA, WPA2 

c. Bluetooth

- Short-range wireless communication (10 m) 
- Operates in **2.4 GHz ISM band** 
- Uses **Frequency-hopping spread spectrum (FHSS)**
- Architecture:
 - **Piconet**: 1 Master + 7 Slaves  
 - **Scatternet**: Multiple piconets interconnected

d. WiMAX

- Long-range **broadband wireless access**  
- Frequencies: 2.3, 2.5, 3.5 GHz
- Offers **high-speed Internet** over large areas 

e. Cellular Telephony

- Used in **mobile phone networks** 📱
- Network is divided into **cells** (हर cell में एक base station होता है)
- Supports:
 - **Handoff** (एक tower से दूसरे में shift)
 - **Frequency Reuse** 🔄
 - Access techniques: FDMA, TDMA, CDMA

f. Satellite Networks

- Used where wired/wireless not possible 🌐 📡
- Types: GEO (High orbit), MEO (Mid), LEO (Low)
- High latency due to long distance ⌚
- Applications: GPS, TV, remote areas

g. Mobile IP

- Allows devices to **move across networks** with **same IP address** 🚶 🌐
 - Key elements:
 - **Home Agent (HA)** 🏠
 - **Foreign Agent (FA)** 🌐
 - **Mobile Node (MN)** 📱
 - Uses **tunneling** to forward packets 📦
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🧱 2. Physical Layer and Transmission Media

a. Data and Signals

- **Data:** Raw 0s and 1s 📄
- **Signals:** Physical representation of data (Electric/Electromagnetic)
- Types: **Analog & Digital**

b. Digital Transmission

- Digital signals sent over media 🖥️ 📡

- Techniques: NRZ, Manchester encoding
- Needs **synchronization** ⌚

c. Analog Transmission

- Analog signals or digital converted to analog via modulation
- Methods:
 - **AM (Amplitude Modulation)**
 - **FM (Frequency Modulation)**
 - **PM (Phase Modulation)** 📡

d. Bandwidth Utilization

- Efficient use of bandwidth 🗨️ ➡️ 📡
- Techniques:
 - **Multiplexing** (FDM, TDM, WDM)
 - **Spectrum Spreading** (FHSS, DSSS - wireless me use hota hai)

e. Transmission Media

- **Guided (Wired)**: Twisted pair, coaxial, fiber optic 📡 📦 💡
 - **Unguided (Wireless)**: Radio waves, microwaves, infrared 🌐 📊
 - Properties: Bandwidth, interference, attenuation, cost
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🎬 3. Multimedia and QoS

a. Data Types

- Text, image, audio, video, binary files 🎵 📷 🗨️
- **Multimedia**: Combination of these

b. Streaming of Audio/Video

- Types:
 - **Stored streaming** (Netflix) 🟡
 - **Live streaming** (Zoom, YouTube Live) 📡
- Needs **buffering** and **synchronization** 🌀

- Uses **RTP (Real-time Transport Protocol)**

c. Real-time Interactive Protocols

- Bi-directional communication in real-time 📞
- Protocols:
 - **SIP (Session Initiation Protocol)**
 - **H.323**

d. Quality of Service (QoS)

- Ensures performance for multimedia 🎯
- Parameters:
 - **Bandwidth** 📊
 - **Delay** ⌚
 - **Jitter** 📈
 - **Packet loss** ✖
- Techniques:
 - **Traffic shaping** 📁
 - **Resource reservation** 📁
 - **Priority scheduling** ⬆

QUESTIONS ??????

? 1. Explain the architecture and working of IEEE 802.11 (Wi-Fi).

📌 Answer:

- **IEEE 802.11** is the standard for **Wireless LAN (WLAN)**.
- Operates in **2.4 GHz** and **5 GHz** frequency bands.
- Uses **CSMA/CA** for avoiding collisions 🔄

-  **Architecture:**
 - **Infrastructure Mode:** Devices connect via **Access Point (AP)**.
 - **Ad-hoc Mode:** Devices connect **directly** to each other.
- **MAC Layer** manages access to the shared medium.
- **Security protocols** include:
 - **WEP** (Weak)
 - **WPA/WPA2** (Stronger )

 **Diagram Expected:** Wi-Fi Infrastructure Mode with AP and Stations.

? 2. What is Bluetooth? Explain Piconet and Scatternet.

 **Answer:**

- Bluetooth is a **short-range wireless communication** technology.
- Operates at **2.4 GHz ISM Band**.
- Uses **Frequency Hopping Spread Spectrum (FHSS)**.

 **Piconet:**

- A small network consisting of **1 master** and **up to 7 slaves**.
- Master controls communication. 

 **Scatternet:**

- **Multiple piconets interconnected.**
- A device can act as master in one and slave in another.

 **Diagram Expected:** Piconet and Scatternet with devices.

? 3. Explain the architecture and applications of Cellular Networks.

 **Answer:**

- Cellular system divides area into **cells**, each with a **base station**.
- Key Concepts:
 - **Frequency Reuse:** Same frequency in non-adjacent cells 

- **Handoff:** Switching from one tower to another 🚶
- Access Techniques:
 - **FDMA:** Frequency Division
 - **TDMA:** Time Division
 - **CDMA:** Code Division

📌 **Applications:** Mobile calling, internet access, SMS

📄 **Diagram Expected:** Cell structure with base stations and handoff arrows.

? 4. What is Mobile IP? How does it work?

📌 **Answer:**

- Enables mobile devices to retain **same IP address** even when changing networks.
- Components:
 - **Mobile Node (MN)** – the device 📱
 - **Home Agent (HA)** – in home network 🏠
 - **Foreign Agent (FA)** – in visited network 🌐

📌 **Working:**

1. MN registers with FA.
2. FA informs HA.
3. Packets from internet sent to HA → **Tunneled** to FA → Delivered to MN

📄 **Diagram Expected:** Mobile IP with HA, FA, MN and tunneling.

? 5. Explain different types of Transmission Media.

📌 **Answer:**

◆ **Guided (Wired):**

- **Twisted Pair:** Used in LAN, cheap, low bandwidth
- **Coaxial Cable:** Used in TV networks
- **Fiber Optic:** High speed, long distance, expensive

◆ Unguided (Wireless):

- **Radio waves:** Omni-directional, used in broadcast
- **Microwaves:** Line-of-sight, used in mobile phones
- **Infrared:** Short range (like TV remote)

 **Diagram Expected:** Comparison of all media types.

? 6. Define Quality of Service (QoS). What are its parameters?

Answer:

- QoS ensures **reliable transmission of multimedia** over the network.

Parameters:

- **Bandwidth** – Data rate available 
- **Delay** – Time taken to travel 
- **Jitter** – Variation in delay 
- **Packet Loss** – Lost packets during transmission 

Techniques:

- **Traffic shaping**
- **Resource reservation**
- **Priority scheduling**

 **Diagram Expected:** Buffering effect with jitter and delay explained.

Diagram – Topics:

1. IEEE 802.11 (Wi-Fi) Architecture

- **Infrastructure Mode** with Access Point and multiple devices
- Also show **CSMA/CA** basic working

● 2. Bluetooth Architecture

- **Piconet:** 1 Master + 7 Slaves

- **Scatternet:** Multiple interconnected piconets
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3. Cellular Network Structure

- Cells with **Base Stations**, showing **handoff** and **frequency reuse**
 - Tower-to-tower movement
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4. Mobile IP Architecture

- Show **Home Agent (HA)**, **Foreign Agent (FA)**, **Mobile Node (MN)**
 - **Tunneling** of data from HA to FA
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5. Types of Transmission Media

- **Guided:** Twisted Pair, Coaxial, Fiber
 - **Unguided:** Radio, Microwave, Infrared
 - Side-by-side visual comparison
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6. QoS Parameters

- Simple graph showing **jitter**, **delay**, **packet loss**, and **bandwidth**
- Buffering vs delay effect